



[25-26_SEM3_CS254_B] - Introduction to Artificial Intelligence

Group 12 Project Proposal

Dr. Daniel Addo

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Team Members:

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AI-Based Academic Performance Prediction System

Group 12

1. Problem Statement and Motivation

Many students fall behind academically before instructors or advisors have a chance to intervene, often because declining grades, reduced engagement, and inconsistent study habits are not identified until it is too late (e.g. after a failed exam or a low midterm grade). Manually tracking these patterns across an entire class or cohort is time-consuming and inconsistent.

This project proposes a system that predicts a student's future academic performance using current and historical data such as grades, attendance, and study habits. The goal is to flag at-risk students early, giving instructors and students themselves the opportunity to intervene (e.g. through tutoring, changed study strategies, or advising) before performance drops significantly. This has clear real-world value for educational institutions seeking to improve retention and outcomes, and gives our team a practical, data-rich problem well suited to applying predictive AI techniques.

2. AI Technique(s) and Rationale

We have identified the following technique as suitable, but may decide to look at other techniques and re-evaluate our selection later on in the project life cycle.

Candidate Technique	Why It's Suitable	Trade-offs / Risks
Random Forest	Works well with different types of data (grades, attendance, habits) without much cleanup. Handles missing/messy data well. Shows which factors matter most (e.g. attendance vs grades), which is easy to explain to instructors. Good for small class-sized datasets.	Can overfit if the dataset is too small. Not as easy to explain as a single decision tree. Needs enough student records to actually learn patterns.

3. Data Sources and Tools Needed

Data sources: Kaggle

Tools and technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab, GitHub

4. Team Roles

Team Member	Role	Primary Responsibilities
Rafiatou	Project Manager	Keeps timeline on track, writes/compiles the report, prepares the final presentation or demo, coordinates team meetings
Fredda	Evaluation & Testing Lead	Tests model accuracy, checks for overfitting/bias
Omuwa	Data Engineer	Finds/collects the dataset, cleans and preprocesses data (handling missing values, formatting grades/attendance), documents data sources
Mitchelle	AI Engineer	Builds and trains the AI model (e.g. Random Forest), handles feature selection, and tunes model performance

5. Milestone Plan (Weeks 10–13)

Week	Milestone / Key Deliverables
Week 10	Collect & clean data. Sketch system design. Set up GitHub repo + README
Week 11	Build baseline model (Random Forest). Commit code regularly.
Week 12	Tune model, start evaluation. Bring a working prototype for the progress demo.
Week 13	Finish evaluation. Run ethics/bias audit. Draft report & slides.